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Tutorial Course for Intro to Sensor-Integrated Systems

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Electrical Loading Effects and Analysis Tools

Core Concept: Loading Effect

- **Ideal State:** The sensor transmits the measurement signal to the next stage (e.g., a recorder) without any loss.
- **Actual Situation:** When the next-stage instrument is connected as a "load," it **draws current** from the sensor, causing a change in the sensor's output signal (typically a voltage drop).
- **Conclusion:** The loading effect is a significant cause of **systematic error** within measurement systems.

Analysis Tool: Thévenin Equivalent Circuit

- Simplifies complex internal sensor circuitry into: an **Ideal Voltage Source (E_{Th})** + **Series Internal Impedance (Z_{Th})**.

Core Formula Derivation:

$$V_L = E_{Th} \frac{Z_L}{Z_{Th} + Z_L}$$

- V_L : Actual measured load voltage
- E_{Th} : Open-circuit voltage of the sensor (the true signal)
- Z_L : Input impedance of the load (measuring instrument)
- Z_{Th} : Equivalent internal impedance of the sensor

Engineering Insights:

- To make the measured value V_L approach the true value E_{Th} , the following condition must be met: $Z_L \gg Z_{Th}$ (Load impedance must be much greater than the sensor's internal impedance).

Potentiometer Loading Effects and Parameter Design

Review of Potentiometric Sensor Principles:

- Uses a sliding contact to change resistance, achieving the conversion from **displacement to voltage**.

Thévenin Equivalent after Load Connection (Let relative displacement $x = d/d_T$):

- **Open-circuit Voltage:**

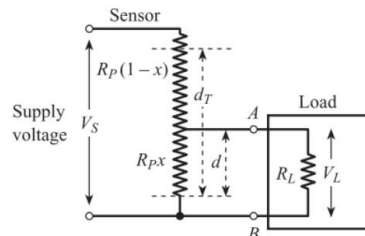
$$E_{Th} = V_S x$$

- **Equivalent Internal Resistance:**

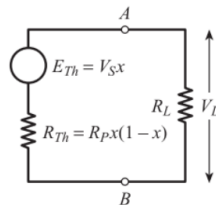
$$R_{Th} = R_P x(1 - x)$$

- **Actual Load Voltage:**

$$V_L = V_S x \frac{1}{(R_P/R_L)x(1 - x) + 1}$$



III



Two Major Constraints in System Design:

1. Non-linear Error

- The connection of a load disrupts the originally perfect linear voltage division.
- **Error Formula:** $\hat{N} \approx 15 \frac{R_P}{R_L} \%$
- **Design Requirement:** Total sensor resistance R_P should be as **small** as possible.

2. Maximum Power Rating

- Sensor heat dissipation must not exceed the limit $\hat{W}$.
- **Power Formula:** $\frac{V_S^2}{R_P} \leq \hat{W}$
- **Design Requirement:** Increasing sensitivity requires increasing the supply voltage V_S , but this is limited by R_P and $\hat{W}$.

Exercise

A linear thermocouple with a sensitivity of $0.04 \text{ mV/}^{\circ}\text{C}$ and resistance of 100Ω is connected to a load with a resistance of $1 \text{ k}\Omega$. Find the voltage across the load for a temperature of 250°C .

Solution

A linear thermocouple with a sensitivity of $0.04 \text{ mV}/^{\circ}\text{C}$ and resistance of 100Ω is connected to a load with a resistance of $1 \text{ k}\Omega$. Find the voltage across the load for a temperature of 250°C .

1. Identify Known Parameters

- Thermocouple Sensitivity (K): $0.04 \text{ mV}/^{\circ}\text{C}$
- Current Measured Temperature (T): 250°C
- Internal Resistance (R_{Th}): 100Ω
- Load Resistance (R_L): $1 \text{ k}\Omega = 1000 \Omega$

2. Detailed Step-by-Step Solution

Step 1: Calculate the theoretical open-circuit voltage (E_{Th})

In an ideal state where no load is connected, the total voltage produced by the thermocouple is determined solely by its sensitivity and the measured temperature:

$$E_{Th} = K \times T$$

Plugging in the values:

$$E_{Th} = 0.04 \times 250 = 10 \text{ mV}$$

Step 2: Calculate the actual load voltage (V_L) using the Voltage Divider Formula

$$V_L = E_{Th} \times \frac{R_L}{R_{Th} + R_L}$$

Substitute the numerical values:

$$V_L = 10 \times \frac{1000}{100 + 1000}$$

$$V_L = 10 \times \frac{1000}{1100}$$

$$V_L = 10 \times \frac{10}{11} \approx 9.09 \text{ mV}$$

Exercise

A potentiometer has a supply voltage of 10 V, a resistance of $10\text{ k}\Omega$ and a length of 10 cm. A recorder of resistance $10\text{ k}\Omega$ is connected across the potentiometer. Calculate the Thévenin equivalent circuit for the sensor and the recorder voltage for each of the following displacements:

- (a) 2 cm
- (b) 5 cm
- (c) 8 cm.

Solution

Known Parameters:

- Supply Voltage (V_S): 10 V
- Total Potentiometer Resistance (R_P): 10 k Ω
- Total Potentiometer Length (d_T): 10 cm
- Recorder (Load) Resistance (R_L): 10 k Ω

Core Formulas (Based on relative displacement $x = d/d_T$):

1. Thévenin Equivalent Voltage (Open-circuit Voltage):

$$E_{Th} = V_S \cdot x$$

2. Thévenin Equivalent Internal Resistance:

$$R_{Th} = R_P \cdot x \cdot (1 - x)$$

3. Actual Measured Load Voltage:

$$V_L = E_{Th} \frac{R_L}{R_{Th} + R_L}$$

(a) When actual displacement $d = 2$ cm:

- Relative displacement (x): $x = 2/10 = 0.2$
- Thévenin Voltage (E_{Th}): $E_{Th} = 10 \times 0.2 = 2.0$ V
- Thévenin Internal Resistance (R_{Th}):
 $R_{Th} = 10 \times 0.2 \times (1 - 0.2) = 10 \times 0.16 = 1.6$ k Ω
- Actual Load Voltage (V_L):

$$V_L = 2.0 \times \frac{10}{1.6 + 10} = \frac{20}{11.6} \approx 1.72 \text{ V}$$

(b) When actual displacement $d = 5$ cm:

- Relative displacement (x): $x = 5/10 = 0.5$ (Center position)
- Thévenin Voltage (E_{Th}): $E_{Th} = 10 \times 0.5 = 5.0$ V
- Thévenin Internal Resistance (R_{Th}): $R_{Th} = 10 \times 0.5 \times (1 - 0.5) = 10 \times 0.25 = 2.5$ k Ω
- Actual Load Voltage (V_L):

$$V_L = 5.0 \times \frac{10}{2.5 + 10} = \frac{50}{12.5} = 4.0 \text{ V}$$

(c) When actual displacement $d = 8$ cm:

- Relative displacement (x): $x = 8/10 = 0.8$
- Thévenin Voltage (E_{Th}): $E_{Th} = 10 \times 0.8 = 8.0$ V
- Thévenin Internal Resistance (R_{Th}): $R_{Th} = 10 \times 0.8 \times (1 - 0.8) = 10 \times 0.16 = 1.6$ k Ω
- Actual Load Voltage (V_L):

$$V_L = 8.0 \times \frac{10}{1.6 + 10} = \frac{80}{11.6} \approx 6.90 \text{ V}$$

Exercise

The motion of a hydraulic ram is to be recorded using a potentiometer displacement sensor connected to a recorder. The potentiometer is 25 cm long and has linear resistance displacement characteristics. A set of potentiometers with maximum power rating of 5 W and resistance values ranging from 250 to 2500 Ω in 250 Ω steps is available. The recorder has a resistance of 5000 Ω and the non-linear error of the system must not exceed 2% of full scale. Find:

- (a) the maximum potentiometer sensitivity that can be obtained;
- (b) the required potentiometer resistance and supply voltage in order to achieve maximum sensitivity.

Solution

1. Organizing Known Conditions and Core Formulas

Known Design Constraints and Parameters:

- Full-scale Displacement (d_T): 25 cm
- Maximum Power Rating ($\hat{W}$): 5 W (Exceeding this will burn the sensor)
- Recorder (Load) Resistance (R_L): 5000 Ω
- Maximum Non-linear Error Requirement ($\hat{N}$): Not to exceed 2% of full scale.
- Available Potentiometer Resistances (R_P): From 250 Ω to 2500 Ω , in increments of 250 Ω (i.e., {250, 500, 750, 1000...}).

Core Formulas:

1. Non-linear Error Limit Formula: $\hat{N} = \frac{400}{27} \frac{R_P}{R_L} \%$ (approximately $15 \frac{R_P}{R_L} \%$)
2. Maximum Power Constraint Formula: $W = \frac{V_s^2}{R_P} \leq \hat{W}$
3. System Sensitivity Formula: $S = \frac{V_s}{d_T}$

Step 1: Determine the usable R_P range based on "Non-linear Error Limits"

To ensure measurement accuracy, the non-linear error must be $\leq 2\%$. Substituting into the limit formula:

$$\frac{400}{27} \frac{R_P}{R_L} \leq 2$$

Solution

Substituting $R_L = 5000 \Omega$:

$$R_P \leq \frac{2 \times 27 \times 5000}{400}$$
$$R_P \leq 675 \Omega$$

Based on the available resistance steps (250 Ω , 500 Ω , 750 Ω ...), the **maximum allowable resistance** is $R_P = 500 \Omega$.

Step 2: Determine the maximum supply voltage V_S based on "Maximum Power Rating"

To maximize sensitivity ($S = V_S/d_T$), we must make the supply voltage V_S as high as possible while d_T remains constant.

Based on the heat dissipation limit:

$$\frac{V_S^2}{R_P} \leq \hat{W} \implies V_S \leq \sqrt{\hat{W} \cdot R_P}$$

This formula clearly shows that **the larger R_P is, the higher the allowable voltage V_S** . This is why we chose 500 Ω instead of 250 Ω in Step 1.

Substituting $\hat{W} = 5 \text{ W}$ and $R_P = 500 \Omega$:

$$V_S^2 \leq 5 \times 500 = 2500$$

$$V_S \leq 50 \text{ V}$$

Thus, the maximum usable supply voltage is $V_S = 50 \text{ V}$.

Step 3: Calculate the maximum system sensitivity

With the maximum voltage and total length known, calculate the sensitivity directly:

$$\text{Max Sensitivity } S = \frac{V_S}{d_T} = \frac{50 \text{ V}}{250 \text{ cm}} = \mathbf{2.0 \text{ V/cm}}$$

Final Answer:

- (a) The maximum achievable potentiometer sensitivity is: 2.0 V/cm
- (b) To achieve this, the required potentiometer resistance is 500 Ω and the supply voltage is 50 V.

Random Signals & Basic Statistical Characteristics

Deterministic vs. Random Signals

- **Deterministic Signals:** Values at future points in time can be accurately predicted (e.g., standard sine waves, step signals).
- **Random Signals:** Interfered with by unknown factors; it is impossible to write a definitive time function $y(t)$ (e.g., actual industrial temperature fluctuations, electromagnetic noise).
- **Strategy:** Give up on predicting "exact values" and instead focus on studying **statistical patterns** over a period of time.

Basic Statistics for Describing Amplitude

- **Mean ($\bar{y}$):** The DC offset component of the signal.
- **Standard Deviation (σ):** The degree of fluctuation/deviation from the mean.
- **Root Mean Square (RMS, y_{rms}):** Measures the energy carried by the signal.
 - *Special Case:* When the mean is 0, the **Standard Deviation = RMS** ($\sigma = y_{rms}$).

Core Energy Formula (Based on 1 Ω impedance reference):

- **Total Power (W_{TOT}):**

$$W_{TOT} = y_{rms}^2$$

Core Analysis Tool: Autocorrelation Function (ACF)

What is Autocorrelation?

- **Physical Meaning:** Measures the degree of "**similarity**" between a signal at the current moment and itself after a **time delay β** .
- **Function:** It acts like a "sieve," filtering out random, chaotic noise to highlight **periodic patterns** hidden within the signal.

ACF Calculation Formula for Discretely Sampled Signals:

$$R_{yy}(m\Delta T) = \frac{1}{N} \sum_{i=1}^N y_i y_{i-m}$$

- N : Total number of sampling points
- m : Number of delay steps ($m \Delta T$ is the total delay time β)
- y_i, y_{i-m} : Sampled values at the current moment and m moments prior.

Analysis of Calculation Steps:

1. Create a duplicate copy of the original signal.
2. Shift the duplicate signal to the right (or left) by m units.
3. **Multiply the aligned values term-by-term**, then **calculate the average** of these products.

Exercise

Two complete periods of a square wave can be represented by the following 20 sample values:

$$+1+1+1+1+1 \quad -1-1-1-1-1 \quad +1+1+1+1+1 \quad -1-1-1-1-1$$

Find the autocorrelation function of the signal over one complete period by evaluating the coefficients $R_{yy}(m\Delta T)$ for $m = 0, 1, 2, \dots, 10$.

Solution

Known Parameters:

- **Signal Waveform:** A square wave (represented by discrete sampling points).
- **Sampling Data:** The problem provides 20 sampling points, representing two full cycles:
+1, +1, +1, +1, +1, -1, -1, -1, -1, -1, +1, +1, +1, +1, +1, -1, -1, -1, -1, -1
- **Single Cycle Length (N):** Since 20 points cover two cycles, one full cycle contains $N = 10$ sampling points.

Core Formula:

For a periodic signal or a sampled signal with a full cycle of length N , the discrete autocorrelation coefficient is calculated as:

$$R_{yy}(m\Delta T) = \frac{1}{N} \sum_{i=1}^N y_i y_{i+m}$$

Meaning: Multiply the original signal (first 10 points) term-by-term with a version of itself shifted left or right by m steps, then sum the products and calculate the average.

We take the original signal sequence for the first cycle ($i = 1$ to 10):

Original Sequence: [+1, +1, +1, +1, +1, -1, -1, -1, -1, -1]

Next, we calculate the R_{yy} value for various delay steps m ($0 \leq m \leq 10$):

① When $m = 0$ (No delay; calculating the signal's total power):

- Original: +1, +1, +1, +1, +1, -1, -1, -1, -1, -1
- Delayed: +1, +1, +1, +1, +1, -1, -1, -1, -1, -1 (0 steps)
- Product: +1, +1, +1, +1, +1, +1, +1, +1, +1, +1
- Sum = 10; Average: $R_{yy}(0) = \frac{10}{10} = 1.0$

② When $m = 1$ (Shifted by 1 sampling period):

- Original: +1, +1, +1, +1, +1, -1, -1, -1, -1, -1
- Delayed: +1, +1, +1, +1, -1, -1, -1, -1, -1, +1
- Product: +1, +1, +1, +1, -1, +1, +1, +1, +1, -1
- Sum = 6; Average: $R_{yy}(\Delta T) = \frac{6}{10} = 0.6$

Solution

③ When $m = 2$:

- Sum = 2; Average: $R_{yy}(2\Delta T) = \frac{2}{10} = 0.2$

④ When $m = 5$ (Delayed by half a cycle):

- Delayed: -1, -1, -1, -1, -1, +1, +1, +1, +1, +1 (Completely out of phase with the original)
- Product: -1, -1, -1, -1, -1, -1, -1, -1, -1, -1
- Sum = -10; Average: $R_{yy}(5\Delta T) = \frac{-10}{10} = -1.0$

⑤ Use symmetry to complete the rest:

Due to the symmetry and periodicity of the ACF for periodic signals, the values will "rebound" linearly after $m = 5$ until $m = 10$, where the waveform overlaps with itself perfectly again ($R_{yy} = 1.0$).

Summary Table of Results:

m	0	1	2	3	4	5	6	7	8	9	10
R_{yy}	1.0	0.6	0.2	-0.2	-0.6	-1.0	-0.6	-0.2	0.2	0.6	1.0

Noise Characteristics & SNR

Internal Noise:

- **Source:** Random thermal motion of electrons within resistors and semiconductors (**Thermal Noise / Johnson-Nyquist Noise**).
- **Characteristics:** Usually manifests as **White Noise**, meaning the energy is distributed uniformly across an extremely wide frequency range.

How to Quantify White Noise? (Core Formulas)

- **Power Spectral Density (PSD, ϕ):** Power per unit frequency (unit: **W/Hz**). Think of this as the "concentration" of the noise.
- **Total Noise Power (W_N):**

$$W_N = \phi \times f_C$$

where f_C is the system bandwidth or cutoff frequency.

- **Noise RMS Voltage ($V_{N,rms}$):**

$$V_{N,rms} = \sqrt{W_N}$$

Assuming a $1\ \Omega$ reference. Since the mean of thermal noise is 0, $V_{N,rms}$ equals the standard deviation σ_N .

Ultimate Evaluation of Signal Quality: Signal-to-Noise Ratio (SNR)

- **Formula:**

$$SNR(dB) = 10\log_{10}\left(\frac{W_S}{W_N}\right) = 20\log_{10}\left(\frac{V_S}{V_N}\right)$$

- **Meaning:** The ratio of useful signal power (W_S) to noise power (W_N).
- **Criterion:** If **SNR < 0 dB**, it indicates that the noise power has exceeded the signal power, and the signal is completely "submerged" or lost in noise.

Interference & Coupling Mechanisms

Interference Sources:

- **50Hz AC Power:** The most common "Mains pick-up" or "Hum."
- **Others:** Transient pulses from motor switching, radio frequency (RF) transmitters, etc.

Coupling Mechanisms:

1. Inductive Coupling: Changes in spatial magnetic fields (di/dt) cut across the measurement loop, inducing an interference voltage.

2. Capacitive Coupling: Distributed "stray capacitance" between high-voltage cables and signal lines allows AC high voltage to leak into the measurement circuit.

3. Multiple Earths / Ground Loops:

- **Phenomenon:** If the sensor and receiver are grounded at different points, a potential difference (V_E) may exist between them, creating an interference current (i_E) in the shield or signal line.

Two Forms of Interference Damage:

- **Series Mode (Normal Mode, V_{SM}):** The interference is superimposed in **series** with the useful signal, leading directly to measurement errors. This is the most critical type.
- **Common Mode (V_{CM}):** The potential of both signal lines rises simultaneously relative to the ground. While it doesn't immediately change the differential voltage between the lines, it can easily convert into series mode interference.

Suppression Techniques for Noise and Interference

Cutting off the Entry Path:

- **Against Inductive Coupling → Twisted Pairs:** Ensures that the induced electromotive forces in adjacent loops have opposite polarities, effectively canceling each other out.
- **Against Capacitive Coupling and Ground Loops → Electrostatic Shielding & Single-point Grounding:** Enclose the circuit in a metal shield and ground it at **only one end** to completely cut off ground loop currents (i_E).
- **Against Common-mode Interference → Differential Amplifier:** Amplifies only the voltage difference between the two lines while ignoring the common-mode voltage that rises simultaneously on both.

•Residual Error Formula:

$$V_{CM_error} = \frac{V_{CM}}{CMRR}$$

where $CMRR$ is the Common-Mode Rejection Ratio; the higher, the better.

Signal Processing Methods (Cleaning Residual Noise):

- **Filtering:** Use a **Band-pass filter** to allow only the signal frequencies to pass through, cutting off noise power outside the bandwidth.

* Signal Averaging:

- **Application:** Periodically repeating signals.
- **Noise Reduction Formula:**

$$\sigma_{AV} = \frac{\sigma}{\sqrt{p}}$$

where p is the number of averages.

- **Principle:** The positive and negative fluctuations of random noise cancel out over multiple averages, while the consistent underlying signal is preserved, significantly improving the SNR.

Exercise

A sinusoidal signal of amplitude 1.4 mV and frequency 5 kHz is 'buried' in Gaussian noise with zero mean value. The noise has a uniform power spectral density of 100 pW Hz^{-1} up to a cut-off frequency of 1 MHz.

- (a) Find the total power, r.m.s. value and standard deviation for the noise signal.
- (b) What is the signal-to-noise ratio in dB?
- (c) Sketch the autocorrelation function for the combined signal and noise.
- (d) The combined signal is passed through a band-pass filter with centre frequency 5 kHz and bandwidth 1 kHz. What improvement in signal-to-noise ratio is obtained?
- (e) The filtered signal is then passed through a signal averager which averages corresponding samples of 100 sections of signal. What further improvement in signal-to-noise ratio is obtained?

Solution

1. Situation Analysis

The Useful Signal

- **Waveform:** Sine wave.
- **Peak Amplitude ($\hat{V}_S$):** 1.4 mV.
- **Frequency (f_S):** 5 kHz.

Gaussian White Noise

- **Mean (μ):** 0.
- **Power Spectral Density (ϕ):** $100 \text{ pW/Hz} = 100 \times 10^{-12} \text{ W/Hz}$ (The "concentration" of the noise).
- **Cutoff Frequency / Bandwidth (f_C):** $1 \text{ MHz} = 10^6 \text{ Hz}$ (The noise covers a very wide frequency range).

Note: System power is calculated based on a 1Ω reference impedance, i.e., $W = V_{rms}^2$

(a) Stage 1: Assessing the Enemy's Damage (Noise Power & Standard Deviation)

1. **Total Noise Power (W_N):** Noise "concentration" multiplied by the "coverage area".

$$W_N = \phi \times f_C = (100 \times 10^{-12} \text{ W/Hz}) \times (10^6 \text{ Hz}) = 100 \times 10^{-6} \text{ W} = \mathbf{100 \mu W}$$

2. **Noise Root Mean Square ($V_{N,rms}$):**

$$V_{N,rms} = \sqrt{W_N} = \sqrt{100 \times 10^{-6} \text{ V}^2} = 10 \times 10^{-3} \text{ V} = \mathbf{10 \text{ mV}}$$

3. **Noise Standard Deviation (σ_N):** Since the mean is 0, the standard deviation equals the RMS value (10 mV).

Solution

(b) Stage 2: Initial SNR

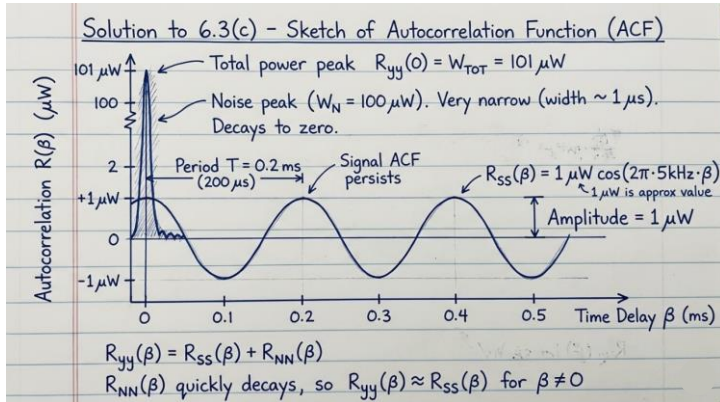
1. **Useful Signal Power (W_S):** Find the RMS value of the sine wave first, then square it.

$$V_{S,\text{rms}} = \frac{\hat{V}_S}{\sqrt{2}} = \frac{1.4 \text{ mV}}{\sqrt{2}} \approx 1.0 \text{ mV}$$

$$W_S = V_{S,\text{rms}}^2 = (1.0 \times 10^{-3} \text{ V})^2 = 1.0 \times 10^{-6} \text{ W} = 1 \mu\text{W}$$

2. **Calculate Initial SNR:**

$$\text{SNR (dB)} = 10 \log_{10} \left(\frac{W_S}{W_N} \right) = 10 \log_{10} \left(\frac{1 \mu\text{W}}{100 \mu\text{W}} \right) = 10 \log_{10}(0.01) = -20$$



Solution

(d) Stage 4: Band-pass Filtering

The noise covers 1 MHz, but our signal is only at 5 kHz. We can use a band-pass filter centered at 5 kHz with a bandwidth (B_{new}) of 1 kHz to "cut away" extra noise.

- **New Noise Power (W'_N):**

$$W'_N = \phi \times B_{\text{new}} = (100 \times 10^{-12} \text{ W/Hz}) \times 1000 \text{ Hz} = 100 \times 10^{-9} \text{ W} = \mathbf{0.1 \mu W}$$

- **New SNR:**

$$\text{SNR}_{\text{new}} = 10 \log_{10} \left(\frac{1 \mu W}{0.1 \mu W} \right) = 10 \log_{10}(10) = \mathbf{10 \text{ dB}}$$

- **Improvement:** $10 \text{ dB} - (-20 \text{ dB}) = \mathbf{30 \text{ dB}}$ (Physically, by narrowing the bandwidth, we eliminated 99.9% of the noise).

(e) Stage 5: Signal Averaging

After filtering, $0.1 \mu W$ of random noise remains. Since the signal is a repeating sine wave, we can use an averager with $p = 100$ cycles.

- **Further Improvement:** Averaging p times reduces random noise power by a factor of p .

$$\text{Further Improvement} = 10 \log_{10}(p) = 10 \log_{10}(100) = \mathbf{20 \text{ dB}}$$

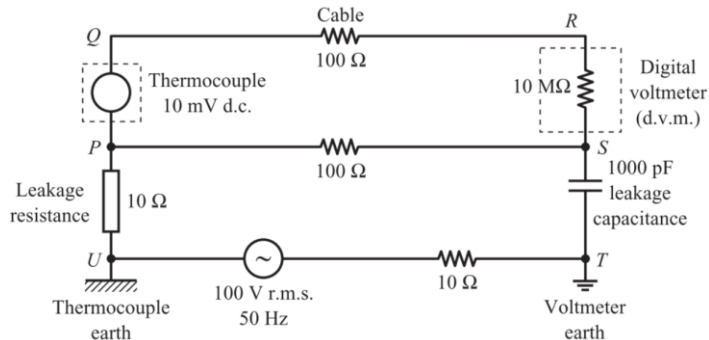
- **Final System SNR:** $10 \text{ dB} + 20 \text{ dB} = \mathbf{30 \text{ dB}}$. The signal power is now 1000 times that of the noise!

Exercise

A thermocouple giving a 10 mV d.c. output voltage is connected to a high impedance digital voltmeter some distance away. A difference in potential exists between earth at the thermocouple and earth at the voltmeter. Using the equivalent circuit given in Figure Prob. 4.

- (a) Calculate the r.m.s. values of the series mode and common mode interference voltages at the voltmeter input.
- (b) If the digital voltmeter has a common mode rejection ratio of 100 dB, find the minimum and maximum possible measured voltages.

Figure Prob. 4.



Solution

1. Site Investigation: Identifying the "Ground Loop"

Background Analysis:

In an industrial setting, a thermocouple (sensor) is grounded at the machine end (Thermocouple earth U), while the digital voltmeter (receiver) is grounded at the control room end (Voltmeter earth T). Due to factors like leakage current between the two locations, an AC potential difference of **100 V (r.m.s.) at 50 Hz** exists in the ground. This creates a closed loop for the **interference current (i_E)**:

- **Path:** Interference source (100 V) $\rightarrow$ 10 Ω leakage resistance $\rightarrow$ Thermocouple terminal $P \rightarrow$ Lower 100 Ω signal cable $\rightarrow$ Voltmeter terminal $S \rightarrow$ 1000 pF leakage capacitance $\rightarrow$ Back to Ground T .

(a) Calculate the RMS values of Series-Mode (V_{SM}) and Common-Mode (V_{CM}) interference voltages

Step 1: Calculate total impedance and current of the interference loop

- At 50 Hz, the capacitive reactance X_C of the 1000 pF leakage capacitor is:

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times (1000 \times 10^{-12})} \approx 3.183 \times 10^6 \Omega = 3.183 \text{ M}\Omega$$

The pure resistance in the loop (10 Ω + 100 Ω + 10 Ω = 120 Ω) is negligible compared to 3.18 M Ω . Therefore, the total loop impedance $|Z_{loop}| \approx X_C$.

- Ground loop interference current (i_E):

$$i_E = \frac{V_E}{|Z_{loop}|} = \frac{100 \text{ V}}{3.183 \times 10^6 \Omega} \approx \mathbf{31.42 \mu A}$$

Solution

Step 2: Calculate Series-Mode Interference Voltage (V_{SM})

Series-mode interference is the error voltage superimposed directly across the signal terminals (between R and S). It is caused by the voltage drop produced by the interference current flowing through the lower cable:

$$V_{SM} = i_E \times R_{\text{cable}} = 31.42 \mu\text{A} \times 100 \Omega = 3142 \mu\text{V} = \mathbf{3.14 \text{ mV (r.m.s.)}}$$

Step 3: Calculate Common-Mode Interference Voltage (V_{CM})

Common-mode interference is the overall voltage lift of the signal lines relative to ground. In this circuit, almost the entire 100 V total voltage is dropped across the component with the largest impedance—the capacitor (from terminal S to ground T):

$$V_{CM} \approx i_E \times X_C = \mathbf{100 \text{ V (r.m.s.)}}$$

(b) Calculate Extreme Values: What will the voltmeter actually display?

Step 1: Conversion of Common-Mode Rejection Ratio (CMRR)

The voltmeter has a CMRR of 100 dB. Converting this to a linear ratio:

$$100 \text{ dB} = 20 \log_{10}(\text{Ratio}) \implies \text{Ratio} = 10^5 = 100,000$$

This means that for a common-mode interference as high as 100 V, only 1/100,000 of it will be converted into harmful series-mode error:

$$V_{CM_error} = \frac{100 \text{ V}}{100,000} = \mathbf{1.0 \text{ mV (r.m.s.)}}$$

Solution

Step 2: Calculate Total AC Interference Peak (V_{Peak})

There are now two AC interference sources at the voltmeter input:

1. **Direct Series-Mode Interference:** 3.14 mV (Resistive voltage drop, in phase with current).
2. **Indirect Series-Mode Error from CM Conversion:** 1.0 mV (Capacitive voltage, lags phase by 90°).

Since the two are 90° out of phase, the total RMS AC interference is calculated using the **root sum of squares**:

$$V_{AC_{rms}} = \sqrt{3.14^2 + 1.0^2} = \sqrt{9.86 + 1.0} \approx 3.30 \text{ mV (r.m.s.)}$$

Convert this to peak AC voltage (maximum amplitude of fluctuation):

$$V_{AC_{peak}} = 3.30 \text{ mV} \times \sqrt{2} \approx \mathbf{4.66 \text{ mV}}$$

Step 3: Determine Measurement Extremes

The true DC signal from the thermocouple is 10 mV. The voltmeter reading will fluctuate based on this 10 mV plus the superimposed AC interference peak:

- **Maximum Measured Voltage** = 10 mV + 4.66 mV = **14.66 mV**
- **Minimum Measured Voltage** = 10 mV - 4.66 mV = **5.34 mV**

Time vs. Frequency Domain

The Limitations of the Time Domain:

- The horizontal axis represents **time**, and the vertical axis represents **amplitude** (similar to the waveform seen on an oscilloscope).
- **Disadvantage:** When useful signals, high-frequency interference, and white noise are all superimposed, the time-domain waveform looks like a tangled mess, making it impossible to distinguish "who is who."

The "Dimensionality Strike" of the Frequency Domain:

- The horizontal axis represents **frequency**, and the vertical axis represents **amplitude/energy**.
- **Core Idea (Fourier's Great Discovery):** Any complex real-world signal can be viewed as the superposition of countless pure sine/cosine waves of different frequencies and amplitudes!
- **Advantage:** Signals that are hopelessly entangled in the time domain will automatically "line up" in the frequency domain based on their frequency levels.

Analogy (For easy memorization):

- The **Time Domain** is like a **cooked dish**: You can only taste the overall flavor (the aggregate waveform).
- The **Frequency Domain** is like the **recipe and ingredient list**: It clearly tells you that the dish contains 10 g of salt (low-frequency signal), 0.5 g of pepper (high-frequency noise), and a pinch of random impurities (white noise).

Core Mathematical Tool: Fast Fourier Transform (FFT)

Discrete Fourier Transform (DFT): Transitioning Digital Signals to the Frequency Domain

➤ Formula:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} kn}$$

- **Meaning:** $x(n)$ represents the discrete data points sampled by the computer; $X(k)$ represents the calculated frequency spectrum data. Through complex exponential operations, it "projects" time-domain data onto various frequency coordinates.

Fast Fourier Transform (FFT):

- Engineers don't need to do the math by hand! FFT is an extremely efficient computer algorithm for calculating the DFT.
- In data-driven analysis software like **Python** or **MATLAB**, a single line of code (e.g., `fft(y)`) can instantly transform tens of thousands of time-domain data points into a spectrum.

How to Read an FFT Spectrum (Frequency Response)?

- **Spikes:** These represent a strong periodic signal existing at that specific frequency. The horizontal coordinate of the spike corresponds to the frequency, while the vertical coordinate represents the amplitude of that signal.
- **Noise Floor:** Because Gaussian white noise contains all frequencies, it doesn't form sharp spikes. Instead, it appears as a "fuzzy" layer of grass spread across the bottom of the spectrum.

The Prerequisite for Digitization: Nyquist Sampling Theorem

Why do we need sampling?

- Computers do not recognize continuous analog waveforms; they can only take discrete "snapshots" at a specific frequency (interval).
- **Sampling Frequency (f_s):** How many points are captured in one second (Units: Hz or Samples/s).

Nyquist-Shannon Theorem (Core Formula):

- To perfectly reconstruct the frequency components of the original signal in a computer without distortion, the sampling frequency must be **strictly greater than twice** the highest frequency component (f_{max}) in the signal.
- **Formula:**

$$f_s > 2f_{max}$$

What happens if the requirement is not met? (Aliasing)

- If you capture data too slowly ($f_s < 2f_{max}$), high-frequency signals will "disguise" themselves as low-frequency signals in the spectrum, leading to a complete analysis error.
- **Example:** In real life, when looking at a fast-rotating car wheel, it sometimes appears to be spinning backward. This is a **visual aliasing** effect caused by the human eye's insufficient sampling rate.

Exercise

Problem 6.6

1. Create a signal $10\sin(2\pi t)$ and plot it.
2. Add a noise signal $0.5\sin(2000\pi t)$ to the original signal and plot it.
3. Add Gaussian noise with a mean of 0 and standard deviation of 1 to the combined signal and plot it.
4. Perform an FFT on the synthesized signal to obtain its frequency response and plot it.

Solution

Problem 6.6

1. Create a signal $10\sin(2\pi t)$ and plot it.
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